

1. Create one VPC in europe (paris)

The screenshot shows the AWS VPC console for the 'Europe (Paris)' region. The breadcrumb navigation is 'VPC > Your VPCs > vpc-0a554add9795b0607'. The left sidebar shows the 'VPC dashboard' and 'Virtual private cloud' section with 'Your VPCs' selected. The main content area displays the details for VPC 'vpc-0a554add9795b0607 / auto scaling project'. The details are organized into four columns:

Details	State	Block Public Access	DNS hostnames
VPC ID vpc-0a554add9795b0607	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-07cf49862307c68cf	Main route table rtb-0683e4e63ce2766ed
Main network ACL acl-062dddc21c982cc	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 598122633072
Encryption control ID -	Encryption control mode -		

2. Create two subnets: one public subnet and one private subnet.

The screenshot shows the AWS VPC console for the 'Europe (Paris)' region, specifically the 'Subnets' page. The breadcrumb navigation is 'VPC > Subnets'. A green notification bar at the top states: 'You have successfully created 1 subnet: subnet-0092f7e20dea4a156'. The left sidebar shows the 'VPC dashboard' and 'Virtual private cloud' section with 'Subnets' selected. The main content area displays 'Subnets (1)' with a search bar and a table of subnets. The table has columns for Name, Subnet ID, State, and VPC. One subnet is listed: 'public scaling' with Subnet ID 'subnet-0092f7e20dea4a156', State 'Available', and VPC 'vpc-0a554add9795b0607 / auto..'. Below the table, there is a 'Select a subnet' section.

aws [Search] [Alt+S] Europe (Paris) shujathuddin (598122633072) shujathuddin

VPC > Subnets

VPC dashboard

Filter by VPC

AWS Global View

▼ Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

You have successfully created 1 subnet: subnet-00a24a492353ed0e0

Subnets (1) Info

Last updated less than a minute ago

Find subnets by attribute or tag

Subnet ID : subnet-00a24a492353ed0e0 Clear filters

☐	Name	Subnet ID	State	VPC
☐	private subnet	subnet-00a24a492353ed0e0	Available	vpc-0a554add9795b0607 auto..

Select a subnet

3. Attach an IGW to the VPC.

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VPC > Internet gateways > igw-00830e719a405a6af

VPC dashboard

Filter by VPC

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Internet gateway igw-00830e719a405a6af successfully attached to vpc-0a554add9795b0607

igw-00830e719a405a6af / auto scaling

Actions

Details Info

Internet gateway ID igw-00830e719a405a6af	State Attached	VPC ID vpc-0a554add9795b0607 auto scaling project	Owner 598122633072
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Tags (1)

Search tags

Key	Value
Name	auto scaling

4. Create one public route table (RT) and one private route table.

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VPC > Route tables > rtb-0c98e01167f0cebb6

VPC dashboard

Filter by VPC

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Elastic IPs

Route table rtb-0c98e01167f0cebb6 | public scaling was created successfully.

rtb-0c98e01167f0cebb6 / public scaling

Actions

Details Info

Route table ID rtb-0c98e01167f0cebb6	Main No	Explicit subnet associations -	Edge associations -
VPC vpc-0a554add9795b0607 auto scaling project	Owner ID 598122633072		

Routes (1)

Filter routes

Both Edit routes

aws [Search] [Alt+S] Europe (Paris) shujathuddin (598122633072) shujathuddin

VPC > Route tables > rtb-08adaeb6b40b25fa2

VPC dashboard < Filter by VPC

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DHCP option sets

Elastic IPs

Managed prefix lists

rtb-08adaeb6b40b25fa2 / private scaling

Details Info

Route table ID
rtb-08adaeb6b40b25fa2

Main
No

Explicit subnet associations
subnet-00a24a492353ed0e0 / private subnet

Edge associations
-

VPC
vpc-0a554add9795b0607 | auto scaling project

Owner ID
598122633072

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (1)

Filter routes

Both Edit routes

Destination Target Status Propagated Route Origin

5. Deploy a NAT gateway in the public subnet and attach the NAT gateway to the private subnet.

VPC > NAT gateways

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

NAT gateways (1) Info

Find NAT gateways by attribute or tag

Create NAT gateway

1

Name	NAT gateway ID	Connectivity...	State	State message	Avai
public scaling	nat-12833147b46515ac7	Public	Available	-	Regi

Select a NAT gateway

VPC > Route tables > rtb-08adaeb6b40b25fa2

Your VPCs

Subnets

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Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

rtb-08adaeb6b40b25fa2 / private scaling

Details Info

Route table ID
rtb-08adaeb6b40b25fa2

Main
No

Explicit subnet associations
subnet-00a24a492353ed0e0 / private subnet

Edge associations
-

VPC
vpc-0a554add9795b0607 | auto scaling project

Owner ID
598122633072

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Filter routes

Both Edit routes

Destination Target Status Propagated Route Origin

6. Create two instances, one in the public subnet and one in the private subnet.

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EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- ▼ Instances
 - Instances
 - Instance Types
 - Launch Templates

Instances (2) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Filter Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	public scaling	i-0dfb48de44f32e585	Running	t3.micro	3/3 checks passed	eu-west-3a
☐	private scaling	i-07f81fe8be0041205	Running	t3.micro	Initializing	eu-west-3a

7. Deploy Apache server on both EC2 instances with a sample index.html file.

Not secure 13.38.26.40

Apache Public Server

This page is running on the public EC2 instance.

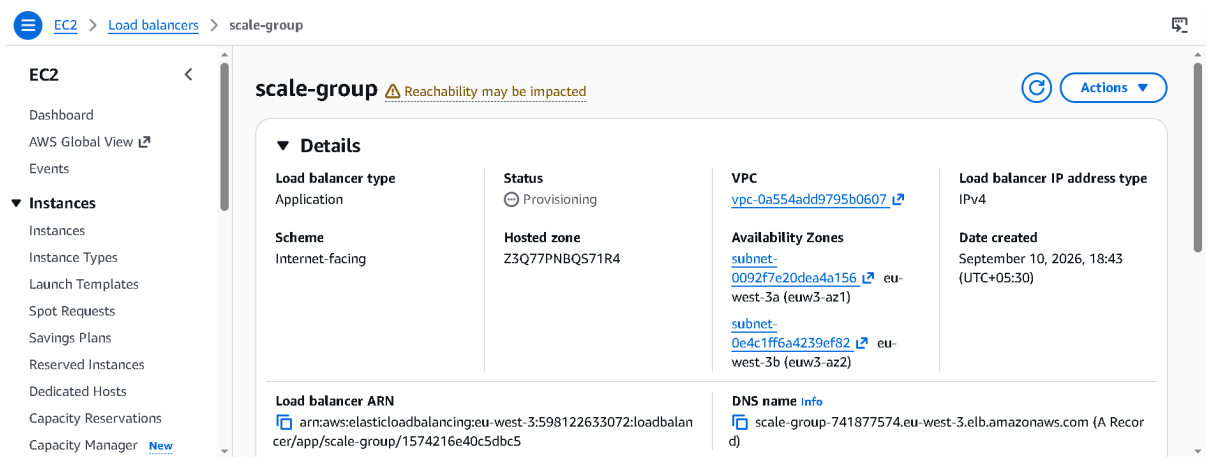
```
[ec2-user@ip-10-0-1-84 ~]$ timeout 5 bash -c '</dev/tcp/10.0.2.108/22' && echo "PORT 22 OPEN" ||
PORT 22 OPEN
[ec2-user@ip-10-0-1-84 ~]$ ls -l ~/privatescaling.pem
ls: cannot access '/home/ec2-user/privatescaling.pem': No such file or directory
[ec2-user@ip-10-0-1-84 ~]$ exit
logout
Connection to 13.38.26.40 closed.

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
13.38.26.40:/home/ec2-user/escalping.pem ~/Downloads/privatescaling.pem ec2-user@1
privatescaling.pem

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ ssh -i ~/Downloads/privatescaling.pem ec2-user@13.38.26.40
#_
~\##### Amazon Linux 2023
~~\#####
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~'~>
~~~
~~~.~.~
~~~/_m/'~>
Last login: Thu Sep 10 12:13:44 2026 from 103.203.172.230
[ec2-user@ip-10-0-1-84 ~]$ ls -l ~/privatescaling.pem
-rw-r--r--. 1 ec2-user ec2-user 1678 Sep 10 12:23 /home/ec2-user/privatescaling.pem
[ec2-user@ip-10-0-1-84 ~]$ chmod 400 ~/privatescaling.pem
[ec2-user@ip-10-0-1-84 ~]$ ssh -i ~/privatescaling.pem ec2-user@10.0.2.108
The authenticity of host '10.0.2.108 (10.0.2.108)' can't be established.
ED25519 key fingerprint is SHA256:dFUVgFaoS1eML8gY13HLz2wkEMi07FBj8D+h1DyTCuM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.2.108' (ED25519) to the list of known hosts.
#_
~\##### Amazon Linux 2023
~~\#####
~~\###|
~~\#/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~'~>
~~~
~~~.~.~
~~~/_m/'~>
[ec2-user@ip-10-0-2-108 ~]$
```

```
[ec2-user@ip-10-0-2-108 ~]$ curl http://localhost
<html>
<head>
<title>Private Apache Server</title>
</head>
<body>
<h1>Apache Private Server</h1>
<p>This page is running on the private EC2 instance.</p>
</body>
</html>
[ec2-user@ip-10-0-2-108 ~]$ exit
logout
Connection to 10.0.2.108 closed.
[ec2-user@ip-10-0-1-84 ~]$ curl http://10.0.2.108
<html>
<head>
<title>Private Apache Server</title>
</head>
<body>
<h1>Apache Private Server</h1>
<p>This page is running on the private EC2 instance.</p>
</body>
</html>
[ec2-user@ip-10-0-1-84 ~]$
```

8. Create one application load balancer and attach it to both EC2 instances.



The screenshot shows the AWS Management Console interface for an Application Load Balancer named "scale-group". The console is in the "EC2" region, and the "Load balancers" page is selected. The "scale-group" load balancer is in the "Provisioning" status. The details section shows the following information:

Load balancer type	Status	VPC	Load balancer IP address type
Application	Provisioning	vpc-0a554add9795b0607	IPv4

Scheme	Hosted zone	Availability Zones	Date created
Internet-facing	Z3Q77PNBQS71R4	subnet-0092f7e20dea4a156 eu-west-3a (euw3-az1) subnet-0e4c1ff6a4239ef82 eu-west-3b (euw3-az2)	September 10, 2026, 18:43 (UTC+05:30)

Load balancer ARN	DNS name
arn:aws:elasticloadbalancing:eu-west-3:598122633072:loadbalancer/app/scale-group/1574216e40c5dbc5	scale-group-741877574.eu-west-3.elb.amazonaws.com (A Record)

EC2 > Target groups > autoscaling-test

autoscaling-test Actions

Details
 arn:aws:elasticloadbalancing:eu-west-3:598122633072:targetgroup/autoscaling-test/7d016c7159afb203

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-0a554add9795b0607
IP address type IPv4	Load balancer scale-group		

2 Total targets	2 Healthy 0 Anomalous	0 Unhealthy	0 Unused	0 Initial	0 Draining
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Distribution of targets by Availability Zone (AZ)
 Select values in this table to see corresponding filters applied to the Registered targets table below.

← ↻ ⚠ Not secure scale-group-741877574.eu-west-3.elb.amazonaws.com

🔖 Import favorites | 🏠 Jira | 📺 Console Home | C... | 🐙 GitHub | 📄 Untitled document...

Apache Public Server

This page is running on the public EC2 instance.

9. Store application load balancer logs in S3.

Amazon S3 > Buckets > scale-group-alb-logs-2026-0910-syed

scale-group-alb-logs-2026-0910-syed info

Objects Metadata Properties Permissions Metrics Management File systems Access Points

Objects (1) 🔄 Copy S3 URI Copy URL Download Open Delete Actions Create folder Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

🔍 Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	AWSLogs/	Folder	-	-	-

10. Store the VPC flow logs in a CloudWatch log group.

VPC

Your VPCs

vpc-0a554add9795b0607

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

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Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

acl-062dddddcd21c982cc

No

10.0.0.0/16

-

IPv6 CIDR

-

Network Address Usage metrics

Disabled

Route 53 Resolver DNS Firewall rule groups

-

Owner ID

598122633072

Encryption control ID

-

Encryption control mode

-

Resource map

CIDRs

Flow logs

Tags

Related resources

Integrations

Flow logs (1)

Actions

Create flow log

Search

Name

Flow log ID

Traffic type

-

fl-05a06c34e11299c09

All

cl